Steps Moving Forward

Goal: build a model to predict non-accrual in borrower loans.

Team: Olivia, Alex, Mincy, Max, Young

Unit of observation: borrower-quarter (loans pooled across BDCs).

Outcomes

1. Non-accrual within the next quarter (binary).
2. Markdown in valuation (continuous), measured two ways: `val_ret` and price change.

Inputs: mark level, rolling max drawdown, PIK switch, valuation dispersion.

Baseline: one simple model per predictor versus each outcome. Logistic regression for non-accrual; simple linear regression for the two markdown outcomes (logistic does not fit a continuous response).

Due next week

- ☐ Agree on the outcome definitions (non-accrual next quarter; `val_ret` and price change) and the borrower-quarter unit
- ☐ Build the four predictors at the borrower-quarter level
- ☐ Fit the baseline models, one predictor at a time, for each outcome
- ☐ Collect the first results (coefficients, fit, AUC for the logits)
- ☐ Draft slides for each presentation topic below

Presentation

Topic	Speaker
The outcome variables being predicted (non-accrual; markdown as `val_ret` and price change)	Max
The choices of models	Alex
The input predictor variables	Young
Results and further exploration	Olivia and Mincy

Model build

1. Define the outcome variable

- ☐ Non-accrual within the next quarter: define the 0/1 flag from the non-accrual data
- ☐ Decide how to treat borrowers already on non-accrual at the start of the quarter
- ☐ Markdown outcomes: `val_ret` (next quarter) and price change (FV/PAR)
- ☐ Confirm the borrower-quarter unit and the sample filters (e.g. minimum prior fair value)

2. Define the input variables

- ☐ Mark level
- ☐ Rolling max drawdown (choose the window)
- ☐ PIK switch (define the switch: when a borrower starts paying in kind)
- ☐ Valuation dispersion (spread of marks across BDCs holding the borrower)
- ☐ Check that every predictor is measured before the outcome quarter, with no look-ahead

3. Select models

- ☐ Baseline: one logistic regression per predictor for non-accrual
- ☐ Baseline: one simple linear regression per predictor for each markdown outcome
- ☐ Combined logistic (non-accrual) and linear (markdown) regression with all four inputs
- ☐ Regularized regression (lasso / ridge)
- ☐ Gradient-boosted trees
- ☐ Neural network (small MLP)

4. Build out

- ☐ Assemble the borrower-quarter panel with outcomes and lagged predictors
- ☐ Split train and test by time, not at random
- ☐ Fit the models and save the outputs

5. Evaluation

- ☐ Classification: AUC and a confusion table at a stated threshold
- ☐ Regression: R^2 and out-of-sample error
- ☐ Compare predictors against each other and against the baseline
- ☐ Note the limits (rare events, overlapping quarters, sample size)